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Syndication, interconnectedness, and systemic risk

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ABSTRACT

Syndication increases the overlap of bank loan portfolios and makes them more vulnerable to contagious effects. We develop a novel measure of bank interconnectedness using syndicated corporate loan portfolios, overlap based on industry and region, and different weights such as equal weights, size and relationships. We find that interconnectedness is driven mainly by bank diversification, less by bank size or overall loan market size. Interconnectedness is positively correlated with different bank-level systemic risk measures including SRISK, DIP and CoVaR, and such a positive correlation mainly arises from an elevated effect of interconnectedness on systemic risk during recessions. Overall, our results highlight that institution-level risk reduction through diversification ignores the negative externalities of an interconnected financial system.

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1. Introduction

The financial crisis of 2007–2009 demonstrated how large risk spillovers among financial institutions caused a global systemic crisis and worldwide economic downturn. The collapse of the interbank market at the beginning of the crisis suggests that direct linkages between banks are an important channel of contagion across financial institutions (Allen and Gale, 2000; Allen and Babus, 2009; Gorton and Metrick, 2012; Giglio, 2016). A second channel that explains how shocks propagate through financial systems is information contagion (Chen, 1999). A third important channel is commonality of asset holdings. As banks have similar exposure to assets such as syndicated loans, a decline in asset prices can affect the banking system, because of direct exposure of banks to the same assets as well as fire sale externalities (e.g. Shleifer and Vishny, 1992, 2011; Kiyotaki and Moore, 1997). Common exposures of banks are of first order importance as indicated by Federal Reserve Chairman Bernanke in his speech at the Conference on Bank Structure and Competition in May 2010 in Chicago[1]:

> *"We have initiated new efforts to better measure large institutions' counterparty credit risk and interconnectedness, sensitivity to market risk, and funding and liquidity exposures. These efforts will help us focus not only on risks to individual firms, but also on concentrations of risk that may arise through common exposures or sensitivity to common shocks. For example, we are now collecting additional data in a manner that will allow for the more timely and consistent measurement of individual bank and systemic exposures to syndicated corporate loans."*

In this paper, we study interconnectedness in the form of common exposures among financial institutions examining banks' exposure to large syndicated loans. The syndicated loan market

[1] Common exposures have played an important role in various historical crises: The Savings & Loans crisis in the U.S. in the 1980s was caused by maturity mismatch of the asset and liability side of banks' balance sheets and a shock to (i.e., increase of) interest rates (Ho and Saunders, 1981). The Asian financial crisis in the 1990s was associated with exchange rate risks. The recent crises in Ireland and Spain were associated with a decline in real estate prices. The 2007–2009 financial crisis involved a decline in real estate prices as well as various forms of contagion magnifying the extent of the crisis (Hellwig, 1995, 2014).

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provides an ideal laboratory to study interconnectedness of banks. It is the most important funding source for non-financial firms (Sufi, 2007), and banks repeatedly participate in syndicated loans arranged by one another. We know borrower and lender identities and are thus able to track banks' investments in this market to quantify common risk exposures.

To measure commonality in banks' syndicated loan portfolio, we develop a novel measure of interconnectedness for which the key component is the similarity between two banks' syndicated loan portfolios.[2] The similarity is measured as the Euclidean distance between two banks based on their exposures to specific borrower industries or regions in the prior twelve months. We then aggregate the distance of one bank with all other banks in each month and construct our bank-level interconnectedness measure using three different weighting schemes: (1) equal weights for each bank, (2) size weights to account for the fact that larger banks might contribute more to systemic risk, and (3) relationship weights to capture prior contractual relationships between banks. Equal weights are used a benchmark against which we evaluate the effect of size and relationships.

We document a high propensity of bank lenders to concentrate syndicate partners rather than to diversify them, as lead arrangers are more likely to collaborate with banks with similar corporate loan portfolios. We then investigate the determinants of interconnectedness both cross-sectionally and over time. While bank size explains only between 5% and 16% of the variation in interconnectedness in the cross-section in univariate tests (depending on the type of exposure and weighting scheme), we document that diversification explains between 61% and 96% of this variation. Overall, our results suggest that bank size is not a first order determinant of interconnectedness but highlights the importance of banks' diversification motive in understanding interconnectedness in the syndicated loan market.

Recent theoretical work has shown that interconnectedness can increase systemic risk through various forms of financial contagion, because of common exposures in times of crises (Allen et al., 2009; Castiglionesi and Navarro, 2010; Ibragimov et al., 2011; Wagner, 2010).[3]

The first channel relies on direct linkages between banks. Once a bank defaults it can propagate stress to other creditor banks (Allen and Gale, 2000).[4] A second important channel is information contagion (Chen, 1999). If one bank is in distress, investors reassess the risk of other institutions that they believe have similar exposures. Short-term investors may decide not to roll over their investments if solvency risks are high, but engage in precautionary liquidity hoarding (Acharya and Skeie, 2011). A third channel is commonality of asset holdings. Shocks can propagate through fire sales, when banks need to sell assets to reduce their leverage. (Shleifer and Vishny, 1992, 2011).[5]

The time-series evolution of our interconnectedness measure is consistent with interpretation of elevated systemic risk through contagion arising from common exposures. We aggregate the bank-level interconnectedness measure to a market interconnectedness index in each month and document that the benchmark equally-weighted interconnectedness index is persistently lower compared to indexes using the size- and relationship-weighting schemes. This is an important finding. For example, the size-weighted index is larger compared to the equally-weighted one which suggests that banks have greater overlap with larger banks consistent with the literature on bank moral hazard and herding behavior (e.g. Acharya and Yorulmazer, 2008) and banks exploiting government guarantees (e.g. Eisert and Eufinger, 2017).

In the final part of the paper, we relate our interconnectedness indexes to different measures of systemic risk. Similar to approaches used in stress tests that have been conducted in the U.S. and Europe since 2008, the construction of these measures is to estimate losses in a systemic stress scenario and determine a bank's equity shortfall after accounting for these losses. These measures capture asset price as well as funding liquidity risks associated with interconnectedness using market data (Acharya et al., 2014).

We employ three frequently used bank-level systemic risk measures: (1) systemic capital shortfall (SRISK) (Acharya et al., 2017; Brownlees and Engle, 2017), (2) distressed insurance premium (DIP) (Huang et al., 2009), and (3) conditional value-at-risk (CoVaR) (Adrian and Brunnermeier, 2016).[6] All three concepts measure a co-movement of equity or credit default swap (CDS) prices without the notion of causality, i.e. a bank can contribute to systemic risk of the financial system because it initiates a contagious event or because of its exposure to a common factor. Moreover, all measures are constructed to estimate cross-sectional differences in systemic risk at a point in time.

We find a positive and significant correlation between our interconnectedness measure and SRIK, but only during recessions. A one standard deviation increase in interconnectedness increases SRISK by almost one-third relative to the average SRISK. Intuitively, a large shock to the market amplifies the effect of interconnectedness if banks are more vulnerable during recessions. Similarly, we find that interconnectedness increases DIP, but also only during recessions. The economic magnitude is comparable, i.e. a one standard deviation increase in interconnectedness increases DIP by about one-third. Bank size is an important determinant of both SRISK and DIP.

We also find that interconnectedness is positively related to CoVaR during recessions. In contrast to the effect of interconnectedness on SRISK, the effect is somewhat smaller. A possible reason is that CoVaR measures the increase in systemic risk of the market when an individual bank is in distress. During recessions, when the market is already weak, the marginal impact of an increase in bank risk is small.

Overall, our bank level tests suggest a positive and significant correlation between our interconnectedness measure and various systemic risk measures including SRISK, DIP, and CoVaR.[7] Controlling for bank and loan market size as well as various fixed effects we show that, consistent with the theoretical papers cited above, interconnectedness amplifies systemic risk during recessions when asset commonality can cause various forms of contagion such as fire-sales.[8] Another way of interpreting this result is that interconnectedness of banks – that builds up during normal times – is a

[2] For example, Abbassi et al. (2017) apply our distance measure to German banks lending portfolios to explain market-based risk measures.

[3] Wagner (2010) shows that diversification increases systemic risk also in the absence of contagion. While diversification reduces the risk of failure of an individual bank, it also increases the likelihood that they default jointly. Moreover, banks can diversify not only in different industries and regions, but also in different sectors (Acharya et al., 2006) such as sovereign debt or household debt which we cannot do due to data limitations.

[4] Allen and Babus (2009), Diebold and Yilmaz (2014), Gorton and Metrick (2012), Duffie (2013) and Giglio (2013) provide further discussions.

[5] Fire-sale amplifications are also discussed in, for example, Kiyotaki and Moore (1997), Brunnermeier and Pedersen (2009), Allen et al. (2012a) and Greenwood et al. (2015).

[6] Other market-based measures (e.g., based on stock return volatility) are developed in Diebold and Yilmaz (2014, 2015), Billio et al. (2012) and Hautsch et al. (2015).

[7] We also show in an Online Appendix A positive and significant link between our interconnectedness measure and the market based CATFIN measure developed by Allen et al. (2012b).

[8] In contrast, Sedunov (2016) proxies a bank's interconnectedness with aggregate measures of loans and derivative positions to other financial institutions – without distinguishing between recession and expansion periods – and finds no effect of interconnectedness on bank-level systemic risk measures.

useful tool to forecast cross-sectional differences in banks' contribution to systemic risk if a severe crisis occurs. Various tests suggest that our results are consistent across different systemic risk measures and model specifications. Consequently, our results highlight that institution-level risk reduction through diversification ignores the negative externalities of an interconnected financial system.

While syndicated loan market exposures reflect, on average, only a percentage of banks' balance sheets, large exposures are of systemic relevance for several reasons. First, the syndicated loan market is extremely large with an annual issuance volume of more than $1000 billion in the U.S.; markets in Europe and Asia are large as well. As has been documented earlier (e.g., Sufi, 2007), virtually all publicly listed firms use this market to finance working capital and for other corporate or capital structure related purposes. Second, individual loans are quite sizeable, too, with respect to a bank's asset size and equity capital.[9] Moreover, banks are financed to some extent with short-term debt instruments and therefore face the risk of bank runs. Additionally, banks' behavior in the syndicated loan market can be used to estimate their systemic risk preferences (Gong and Wagner, 2016). Banks want to be correlated with other institutions and interconnectedness in the syndicated loan market can be used to measure these preferences that extend to other asset classes of banks' balance sheets.[10]

The paper proceeds as follows. In Section 2, we describe the empirical methodology, in particular, derive our measures of distance and interconnectedness, and discuss various systemic risk measures as well as the related literature. Data are described in Section 3. Sections 4 and 5 discuss our empirical results on interconnectedness in loan syndications and the implications of such interconnectedness for systemic risk. Finally, we conclude in Section 6 with some policy implications.

2. Empirical methodology

In this section, we first develop our interconnectedness measure and then briefly describe the different systemic risk measures used in our empirical tests. All variables are defined in Table A1.

2.1. Measuring interconnectedness

In this subsection, we describe how we measure distance between two banks based on lending specializations. We then explain how we construct our interconnectedness measure.

2.1.1. Distance between two banks

We analyze bank syndicated loan specialization related to U.S. borrower industry and borrower geographic locations. Specifically, we use the 2-digit borrower's SIC industry code and the borrower's U.S. state in which it has its headquarter to examine in which area(s) each bank has heavily invested.[11] Analyzing specializations along industries and regions captures two key diversification dimensions of banks' risk management.[12] We then compute the distance between two banks by quantifying the similarity of their loan portfolios. The detailed construction of our distance measure is as follows.

For each month during the January 1989 to June 2011 period, we compute each lead arranger's total loan facility amount it originated during the prior 12 months, using Dealscan's loan origination data.[13] There were approximately 70–125 active lead arrangers each month in the U.S. syndicated loan market; as a result, we obtain 27,117 unique lead arranger-months. We then compute portfolio weights for each lead arranger in each specialization category (2-digit borrower SIC industry, borrower headquarter U.S. state). Let $w_{i,j,t}$ be the weight lead arranger i invests in specialization (e.g. industry) j within 12 months prior to month t. Note that for all pairs of i and t, $\sum_{j=1}^{J} w_{i,j,t} = 1$, where J is the number of industries, or geographic locations the lender can be specialized in.

Next, we compute the distance between two banks as the Euclidean distance between them in this J-dimension space:

$$\text{Distance}_{i,k,t} = \frac{1}{\sqrt{2}} \times \sqrt{\sum_{j=1}^{J} \left(w_{i,j,t} - w_{k,j,t}\right)^2}, \quad (1)$$

where $\text{Distance}_{i,k,t}$ is the distance between bank i and bank k in month t ($i \neq k$). The distance measure is normalized to a scale of 0–1 with 0 reflecting no distance (complete portfolio matching) and 1 reflecting full distance (i.e. no portfolio overlap). Appendix B provides an example on how distance between two banks is computed as specified in (1). For illustrative purposes, we discuss the computation of our distance measures based on borrower SIC industry for JPMorgan Chase, Bank of America, and Citigroup. These three banks were the top three lead arrangers as of January 2007 according to their portfolios of syndicated loans originated during the prior twelve months (i.e. January-December 2006). Citigroup had a different loan portfolio from those held by either JPMorgan Chase or Bank of America, investing more heavily in the manufacturing, transportation, communications, electric, gas, sanitary, and services industries and less heavily in retail trade, finance, insurance and real estate. As a result, the distance computed between Citigroup and either JPMorgan Chase or Bank of America is greater than the distance between JPMorgan Chase and Bank of America whose portfolios were more similar to each other. We show the pairwise distance in Appendix C.

2.1.2. Bank-level interconnectedness

To measure monthly interconnectedness at the bank-level, we first take the weighted average of the distance between a given lead arranger and all the other lead arrangers in the syndicated loan market. As a smaller Euclidean distance means higher interconnectedness, we then linearly transform the weighted average of distance into an interconnectedness measure for the lead arranger.

[9] On average, the ratio of syndicated corporate loans originated during a 12-months period as a percentage of bank total assets is 9.6%. Other factors need to be considered as well: For example, suppose that book equity is about 5% or less relative total assets, then syndicated loans on a bank's balance sheet is eventually twice a bank's book equity.

[10] While new syndicated loan origination declined during the 2007–2008 financial crisis in the US, the total lending exposure on banks' balance sheet increased substantially (Ivashina and Scharfstein, 2010). Borrowers started to draw down credit lines that have been committed by banks during the period of credit expansion. In fact, a large percentage of loans in the Dealscan sample are credit lines (Berg et al., 2016). In other words, the committed amounts of loans that have been originated before the recession period are a good proxy for the portfolio of banks when they enter the recession. We do not differentiate between credit lines and term loans when calculating our interconnectedness measure and our proxies thus captures the drawn credit line exposures.

[11] We also examine lender specialization at different aggregation levels for borrower industry (SIC industry division, 3-digit SIC industries, 4-digit SIC industries) and borrower geographic location (U.S. region, 3-digit zip code). We obtain very similar results.

[12] Note that U.S. borrower geographic location and industry are correlated. Certain industries tend to concentrate in certain areas. For example, WY, WV, LA, OK, and TX have higher concentration in mining, whereas WI, VT, OH, IN, OR, NC, and CT are more specialized in manufacturing. Consequently, we expect borrower geographic location to generate results somewhat similar to those based on borrower industry.

[13] The loan amount is split equally over all lead arrangers for loans with multiple leads.

Our measure is normalized to a scale of 0–100 with 0 being least interconnected and 100 being most interconnected.[14] That is, a higher value indicates a more interconnected bank. Specifically, the interconnectedness of bank i in month t, $\text{Interconnectedness}_{i,t}$, equals:

$$\text{Interconnectedness}_{i,t} = \left(1 - \sum_{k \neq i} x_{i,k,t} \cdot \text{Distance}_{i,k,t}\right) \times 100, \quad (2)$$

where $\text{Distance}_{i,k,t}$ is the distance between bank i and bank k in month t as defined in (1), and $x_{i,k,t}$ is the weight given to bank k in the computation of bank i's interconnectedness.

We use three kinds of weighting schemes for $x_{i,k,t}$: We assign equal weights to all other banks ('equal-weighted interconnectedness'). This is our baseline specification to separate the (incremental) effect of adjusting weights to capture two distinct bank characteristics (size, and lending relationships). Specifically, we construct also a size weight of loan holdings where larger banks may matter more for systemic risk ("size-weighted interconnectedness"). The weights are computed based on the lagged total assets of bank k relative to the sum of lagged total assets of all other banks in the syndicated loan market. Finally, we construct a relationship-weighted measure of interconnectedness to capture the importance of lending relationships among lead arrangers for the overlap of bank's syndicated loan portfolios ("relationship-weighted interconnectedness"). The idea here is to capture implicit relationships in sharing loan syndication among closely related (compared to distantly related) banks.[15] Specifically, the weight is calculated based on the number of collaborative relationships between bank i and bank k relative to the total number of relationships bank i had with all banks in the syndicated loan market during the prior twelve months.[16] Appendix D provides an illustrative example on how all three different weights among banks are computed.

2.1.3. Market-aggregate interconnectedness

We use our individual bank measures to construct monthly Interconnectedness Indexes aggregating bank-level interconnectedness to the market level using a simple average. As we have three weighting schemes on the bank level (equal-, size-, and relationship-weighting), we obtain three monthly Interconnectedness Indexes.

$$\text{InterconnectednessIndex}_t = \sum_i \frac{1}{N_t} \cdot \text{Interconnectedness}_{i,t}, \quad (3)$$

where $\text{Interconnectedness}_{i,t}$ is the interconnectedness of bank i in month t as defined in (2) and N_t is the number of connected banks as of month t.[17]

2.2. Measuring systemic risk

To analyze the link between loan portfolio interconnectedness and systemic risk, we use three bank-level systemic risk measures proposed in the recent literature: (i) SRISK, (ii) DIP, and (iii) CoVaR. These measures are briefly described below.

2.2.1. SRISK

SRISK is a bank's U.S.-Dollar capital shortfall if a systemic crisis occurs, which is defined as a 40% decline in aggregate banking system equity over a 6-month period. This measure is developed in Acharya et al. (2017) and Brownlees and Engle (2017). SRISK is defined as

$$\begin{aligned} \text{SRISK} &= E((k(D + MV) - MV)|\text{Crisis}) \\ &= kD - (1 - k)(1 - \text{LRMES})MV, \end{aligned} \quad (4)$$

where D is the book value of debt which is assumed to be unchanged over the crisis period, LRMES is long-run marginal expected shortfall that measures the co-movement of a bank's stock price with the stock market index when the overall market return falls by 40% over the crisis period.[18] LRMES × MV is then the expected loss in market value of a bank over this 6-month window. k is the prudential capital ratio which is assumed to be 8% for U.S. banks and 5.5% for European banks to account for differences between US-GAAP and IFRS measures. SRISK thus combines both the firm's projected market value loss due to its sensitivity to market movements and its leverage. Naturally, SRISK is larger for larger banks. To make sure our results are not driven solely by bank size, we conduct various tests and carefully control for bank size in our tests.

2.2.2. DIP

We use the DIP as our second market-based measure of systemic risk (Huang et al., 2009, 2012; Black et al., 2013). The four main components of DIP are: (1) the risk-neutral probability of default (PD), which is calculated from CDS prices using (2) loss given default (LGD) estimates, which can vary over time, (3) asset correlations which are measured using equity return correlations, and (4) the total liabilities of all banks.

Huang et al. (2009) construct a hypothetical portfolio of the total liabilities of all banks and use Monte-Carlo simulations to estimate the risk neutral probability distribution of credit losses for that portfolio. DIP is then a hypothetical insurance premium to cover losses if total losses (L) (aggregated over all banks) exceeds a certain threshold of a banks' total liabilities (L_{min}). DIP can then be expressed as follows:

$$\begin{aligned} \text{DIP} &= E^Q(L|L > L_{min}) \\ \frac{\partial \text{DIP}}{\partial L^i} &= E^Q(L^i|L > L_{min}) \end{aligned} \quad (5)$$

DIP describes a conditional expectation of portfolio losses under extreme conditions. It is thus similar to an expected shortfall concept, but is not defined using a percentile distribution but rather by using an absolute loss threshold (L_{min}). In that sense, it is also similar to SRISK.[19] L^i is then the loss of an individual institution and determines the marginal contribution of a bank to the systemic risk of the financial sector $\left(\frac{\partial \text{DIP}}{\partial L^i}\right)$. While we consistently refer to this measure as "DIP" throughout the paper, we operationalize it using the loss of each individual bank in the regressions (i.e., L^i).

[14] We can also interpret an interconnectedness value of 0 as being not interconnected at all (i.e., having a loan portfolio completely different from all the other lead arrangers' portfolios) and 100 as being totally interconnected (i.e., have a loan portfolio exactly same as all the other lead arrangers' portfolios).

[15] We also constructed a fourth weighting scheme using the actual U.S. dollar exposures of banks to the same borrowers ("exposure-weighted interconnectedness"). Our main results of the paper do not change under this fourth weighting scheme.

[16] A collaborative relationship is identified if bank k is bank i's participant lender, co-lead, or lead arranger.

[17] Instead of using a simple average, we could also use e.g. bank size to aggregate the bank-level interconnectedness to the market level. The results do not change and are not reported for brevity.

[18] V-Lab uses the S&P 500 for U.S. banks and the MSCI ACWI World ETF Index for European banks.

[19] The major methodological difference between DIP, SRISK and CoVaR is that DIP is a risk-neutral measure, while SRISK and CoVaR are statistical measures using physical distributions. From an economic perspective, DIP is different compared to shortfall measures such as SRISK as the CDS spreads used to calculate default risk measure the potential losses to debt holders assuming all equity is wiped out. One can therefore also refer to DIP as a "bailout measure," which is quite often the focus in policy discussions of this measure.

2.2.3. CoVaR

Our third market-based measure of systemic risk is CoVaR (Adrian and Brunnermeier, 2016). CoVaR is the value at risk (VaR) of the financial system conditional on one institution being in distress and ΔCoVaR is the marginal contribution of that firm to systemic risk. The VaR of each institution is measured using quantile regressions and the authors use a 1% and 5% quantile to measure CoVaR:

$$\text{Prob}\left(L \geq \text{CoVaR}_q | L^i \geq \text{VaR}_q^i\right) = q, \quad (6)$$

where L is the loss of the financial system, L^i is the loss of institution i, and q is the VaR quantile (for example, 1%). Thus, CoVaR measures spillovers from one institution to the whole financial system. Importantly, CoVaR does not imply causality, i.e., it does not imply that a firm in distress causes the systemic stress of the system, but rather suggests that it could be both, a causal link and/or a common factor (in terms of asset or funding commonality) that drives a bank's systemic risk contribution.

CoVaR is not as sensitive to size or leverage as SRISK. Moreover, in contrast to SRISK, CoVaR includes only the correlation with market return volatility, but not a bank's return volatility. Suppose that two banks have the same market return correlation, but bank A has low volatility while bank B has high volatility. Both banks would have the same CoVaR even though bank A is essentially of low risk.

2.2.4. Comparing the three bank-level systemic risk measures

We use the three different systemic risk measures above, as they capture different aspects of systemic risk. SRISK captures the consequences for an *individual* bank resulting from a 40% equity decline of the entire *aggregated* banking system. DIP (similar to SRISK) captures the loss of an *individual* bank conditional on expectations of the *aggregated* banking sector being under extreme conditions. CoVaR analyzes the reverse. It captures the VaR of the *aggregated* financial system resulting from an *individual* institution being in distress.

These differences might also affect the sensitivities of the systemic risk measures to interconnectedness during both expansions and recessions. Interconnectedness might have a larger impact on SRISK and DIP during recessions. During recessions, banks' capital levels and therefore loss absorbing capacities are lower. This is exactly when banks are most vulnerable, so that higher interconnectedness may facilitate spillovers when the system as a whole is shocked.

The sensitivity of the CoVaR measure is less obvious. In expansions, a large shock of a single institution may lead to less severe spillovers to the whole financial system, but might have a large impact to the sensitivity of VaR tail-risks as vulnerabilities of all other banks rise. In recessions, spillovers should be larger, but given that the whole financial system was just shocked by a common factor, the marginal impact on changes to the VaR tail-risks might be lower. The effect of interconnectedness on CoVaR might thus be lower relative to the other two systemic risk measures.

3. Data and summary statistics

In this section, we discuss our data sources and provide summary statistics.

3.1. Data sources

We use two primary sources of data to analyze the interconnectedness of banks in loan syndication and how such interconnectedness affects banks' systemic risk: (i) syndicated loan data and (ii) systemic risk data. Thomson Reuters LPC DealScan is the primary database on syndicated loans with comprehensive coverage, especially for the U.S. market. We use a sample of 90,578 syndicated loan facilities originated for U.S. firms between January 1988 and June 2011 to construct our distance and interconnectedness measures.[20] These loans present very similar characteristics as documented in the literature, e.g., Sufi (2007) and Ivashina (2009). We also follow that literature identifying the lead arranger of the syndicate.

We measure interconnectedness at the lead arranger (bank holding company) level for different reasons. Lead arrangers establish the relationship with the borrowing firm and decide on the syndication structure of the loan. In addition, lead arrangers take larger shares in the syndicated loan due to information asymmetries between the lead arranger and participants (Sufi, 2007).[21]

We use lead banks, as they also actively participate. Lender arrangements in about seven out of ten syndicated loans are reciprocal in the sense that lead arrangers also participate in loans that are led by their participant lenders (Cai, 2010). Thus, lead arrangers are a source of interconnectedness as they syndicate loans for others to be exposed to their loan portfolio risk, and at the same time, they are exposed to other banks' loan portfolio risk as they participate. This is not the case for banks that are only participants (but are never lead arrangers). These participant banks are passively chosen by lead arrangers in loan syndicates, but do not actively increase interconnectedness with other large lenders in the syndicated loan market.

Importantly, we show in this paper that lead arrangers work together with banks that have similar portfolios measured by loan portfolio allocation as lead arrangers as well as those they have previous relationship based on loan market collaboration. In other words, their allocation as lead arrangers should be close to the portfolio allocation as participants.

Our sample of lead arrangers consists of U.S. institutions as well as large international institutions that have significant syndicated loan exposure originated in the U.S. market.[22] Moreover, 26 of 28 Globally Significant Financial Institutions (G-SIFIs) are included in our sample.

Using lead bank data can thus be a good approximation for loan participation, in particular for loan participation arising from (systemically relevant) reciprocal relationships as lead and participants. As alluded to above, these comprise the major part of syndicated loan participants.

We obtain the SRISK data from NYU V-Lab's Systemic Risk database and the DIP and CoVaR data from the authors who proposed them as systemic risk measures. SRISK data covers 132 global financial institutions and 16,258 bank-months ranging from January 2000 to December 2011. We are able to match them with 5733 lead arranger-months and 58 unique lead arrangers. The DIP data are weekly covering 57 unique European financial institutions from January 2002 to January 2013. We aggregate weekly data into monthly measures and obtain 5235 bank-months with DIP mea-

[20] We have an original sample of 91,715 syndicated loan facilities. However, we restrict ourselves to lead arrangers that originated at least ten syndicated loans in the sample period to exclude lead arranger that only randomly enter the syndicated loan market (these are usually very small banks that should matter less for the interconnectedness of a bank and the banks' systemic risk). In total, our sample consists of 223 lead arrangers. We receive very similar results, if we do not restrict the number of lead arrangers.

[21] Lead arrangers (usually one or two per syndicated loan) each hold, on average, a 30 percent share in the syndicated loan, while participants hold, an average, 8 percent each. Dealscan has limited information about the loan shares retained by banks at origination. The subsample with loan share information comprises about 20 percent of the loan observations.

[22] Non-U.S. financial institutions have large exposures to U.S. borrowers. For example, we document in Appendix C, that five European banks are among the top ten lead arrangers in the U.S. syndicated loan market as of January 2007. Overall, non-U.S. lead arrangers originated 17.4% of the total loan amount in the U.S. syndicated loan market.

sures. We are able to construct a matched sample of 19 unique lead arrangers and 1343 lead arranger-months with our interconnectedness data. The CoVaR data are quarterly covering 1194 public U.S. financial institutions and 62,642 bank-quarters ranging from the first quarter of 1986 to the fourth quarter of 2010. We are able to match them with 1716 lead arranger-quarters and 38 unique lead arrangers.[23] In our regressions, we use CoVaR data starting in the first quarter of 2000 to ensure comparability across our three systemic risk measures when computing the portfolio overlap.

3.2. Summary statistics

Table 1 reports summary statistics for the distance, interconnectedness, and systemic risk measures described in Section 2 as well as lead arranger (bank) and market characteristics. Distance is summarized for approximately 2,700,000 lead arranger pair-months and interconnectedness of roughly 26,500 lead arranger-months across the two lender specialization categories and three interconnectedness weightings. Lender specialization can be by industry and geographic location, while interconnectedness can be equal-, size-, and relationship-weighted. By definition, distance must lie within the range of 0–1 and interconnectedness within the range of 0–100. The standard deviations of these measures imply that there is sufficient variation for empirical tests. Further, the distribution of our distance measures indicate a similar variation of lender specialization differences across banks for both industry and geographic location. This similarity of variations in industry and geographic location specializations carries over to the bank-level and market aggregate interconnectedness measures. However, our different weighting schemes do capture distinct differences in the interconnectedness of lead arrangers. The size-, and relationship-weighted interconnectedness measures are on average much larger than the baseline equally-weighted measure. Furthermore, lending relationships appear to matter more for interconnectedness than size. This result holds for both the bank-level and the market-aggregate interconnectedness measures.

Summary statistics of SRISK, DIP, and CoVaR are reported at the lead arranger level. Our SRISK data consists of 5733 matched lead arranger-months from U.S. and international institutions heavily invested in the U.S. syndicated loan market, and the average SRISK is $25.3 billion. Our DIP data consists of 1343 matched lead arranger-months from European banks heavily invested in the U.S. syndicated loan market, and the average DIP is $20.4 billion converted into U.S. dollar-values based on the exchange rate at beginning of year 2009. Our CoVaR data consists of 1716 matched lead arranger-quarters from U.S. institutions, with the 1% CoVaR equal to a decline of $15.8 billion in bank equity value on average.[24] All these measures show greater systemic risk for our sample of lead arrangers than an "average" financial institution in the SRISK, DIP, and CoVaR data sets.[25] The SRISK measure and DIP measure have correlations close to 0.8 for the sample of lead arrangers for which the data is available. The correlation between SRISK and CoVaR is about 0.2.[26]

[23] The Appendix contains a lists lead arrangers for which the various systemic risk measures are available.

[24] The CoVaR data are all expressed in the form of losses, i.e., negative numbers. In our empirical analyses, we multiply CoVaR with minus one so that a higher CoVaR implies higher systemic risk.

[25] For example, the SRISK of an average financial institution is $10.3 billion. An average public U.S. financial institution in the 1% CoVaR data shows a decline of $0.785 billion, and an average European financial institution in the DIP data shows a DIP of $10.9 billion.

[26] In our sample, the DIP and CoVaR measures do not overlap, as our CoVaR sample comprises U.S. institutions, while our DIP sample includes European banks.

4. Interconnectedness of banks in loan markets

In this section, we first empirically examine how banks become interconnectedness in the U.S. syndicated loan market. We then explore the determinants of interconnectedness and analyze the trend of interconnectedness over time.

4.1. How do banks become interconnected?

To understand commonality of banks' syndicated corporate loan portfolio, we first examine how banks become interconnected in the syndicated loan market. In order to make our data and computations manageable, we limit our interest to the top 100 lead arrangers in each month that hold an aggregated share of at least 99.5% of the total market. We estimate the following regression:

$$\begin{aligned}\text{Syndicate Member}_{i,k,n,t} &= \alpha + \beta_1 \cdot \text{Distance}_{i,k,t} \\ &+ \beta_2 \cdot \text{Lead Relationship}_{i,k,t} + \beta_3 \cdot \text{Borrower Relationship}_{k,n} \\ &+ \beta_4 \cdot \text{Market Share}_{k,t} + \text{Loan Facility}_n' + e_{i,k,n,t}, \end{aligned} \quad (7)$$

where the dependent variable Syndicate Member$_{i,k,n,t}$ is an indicator variable that equals one if lead arranger i chooses lender k as a member in loan syndicate n that is originated in month t and zero otherwise. Distance$_{i,k,t}$ measures the distance between lead arranger i and lender k based on their syndicated loan portfolios during the twelve months prior to month t. As a proxy for bank-to-bank relationships, Lead Relationship$_{i,k,t}$ is an indicator variable for whether lead arranger i had syndicated any loans with lender k prior to the current loan (no matter what roles the two lenders took). As a measure of bank-firm relationships, Borrower Relationship$_{k,n}$ is an indicator variable for whether lender k arranged or participated in any syndicated loans that were made to the borrower prior to loan syndicate n. By including Lead Relationship$_{i,k,t}$ and Borrower Relationship$_{k,n}$ in the regression, we control for the effects of prior relationships between the two lenders and prior relationships between the borrower and lender k on the construction of the syndicate. Market Share$_{k,t}$ is the market share of lender k as a lead arranger during the twelve months prior to month t. We use Market Share$_{k,t}$ to proxy for lender k's reputation and market size or power. Loan Facility$_n$ is a vector of loan facility fixed effects, which are included to rule out any facility-specific effects. Standard errors are heteroscedasticity robust and clustered at the lead arranger level. The resulting sample size is almost 11 million lender pairs.

The results are reported in Table 2. In all regressions, our distance measures show negative coefficients that are significant at the 1% level. That is, the greater the portfolio similarity between a lender and the lead arranger, the greater the likelihood that the lender is chosen as a syndicate member. We also find that a lender's prior relationships with either the lead arranger or the borrower have significantly positive influence on the likelihood of being chosen as a syndicate member. The effect is especially strong for prior lender-borrower relationships, which is consistent with the findings in Sufi (2007). Moreover, lender k's market share increases its likelihood of being included in the syndicate.

Overall, the results suggest that lead arrangers tend to work with banks that have more similar corporate loan portfolios increasing the degree of bank interconnectedness over time.

4.2. Determinants of interconnectedness

To understand the determinants of interconnectedness, we examine the effect of three bank characteristics, (i) total assets, (ii) diversification, and (ii) number of specializations and loan market size. Total assets is a standard proxy for bank size; the other

Table A1
Variable Definitions.

Variable	Definition
Borrower Relationship	An indicator variable for whether a potential lender has previous relationships with the borrower
CoVaR	1% contagion value-at-risk of a U.S. bank measured in billions of U.S. dollars
DIP	Distressed insurance premium of a European bank in billions of euros
Distance	Distance between two banks based on their syndicated loan portfolios as lead arrangers during the previous twelve months
Diversification	Diversification of a bank based on its syndicated loan portfolio
Expansion	An indicator variable for whether a month falls into an expansion period, defined as a month not identified as a recession by the NBER
Interconnectedness	Bank-level interconnectedness
Interconnectedness Index	Market-aggregate interconnectedness
Lead Arranger Fixed Effect	Lead arranger (bank) fixed effect
Lead Relationship	An indicator variable for whether a potential lender has previous relationships with the lead arranger
Loan Facility Fixed Effect	Loan facility fixed effect
Market Share	Market share of a bank in the U.S. syndicated loan market based on the total loan amount the bank originated as a lead arranger
Market Size	The size of the U.S. syndicated loan market measured by the total newly originated syndicated loan amount in billions of U.S. dollars
Number of Specializations	Number of specializations a bank is engaged in as a lead arranger
Recession	An indicator variable for whether a month falls into recessions as identified by the NBER
SRISK	Systemic capital shortfall of a bank measured in billions of U.S. dollars
Systemic Risk	Any systemic risk measure
Syndicate Member	An indicator variable for whether a potential lender is chosen by the lead arranger to be a loan syndicate member
Total Assets	Book value of a bank's total assets in billions of U.S. dollars

This Appendix lists the variables used in the empirical analysis and their definitions.

Table 1
Summary Statistics.

	N=	Mean	SD	10th	50th	90th
Distance Measures:						
Distance in 2-digit Borrower SIC	2,692,389	0.630	0.222	0.320	0.664	0.898
Distance in Borrower State	2,690,674	0.634	0.229	0.313	0.672	0.917
Bank-Level Interconnectedness Measures:						
Equal-weighted Interconnectedness:						
Based on 2-digit Borrower SIC	26,741	36.1	14.0	16.1	37.8	54.0
Based on Borrower State	26,532	35.8	14.2	15.7	37.4	53.6
Size-weighted Interconnectedness:						
Based on 2-digit Borrower SIC	26,741	42.9	16.2	20.6	44.1	64.2
Based on Borrower State	26,532	43.5	16.7	20.3	44.9	65.0
Relationship-weighted Interconnectedness:						
Based on 2-digit Borrower SIC	26,483	46.7	17.2	23.0	48.0	69.1
Based on Borrower State	26,277	46.4	17.1	22.5	47.5	68.2
Market-Aggregate Interconnectedness Measures:						
Equal-weighted Interconnectedness:						
Based on 2-digit Borrower SIC	270	35.7	4.0	29.1	36.8	39.8
Based on Borrower State	270	35.3	4.7	28.3	35.8	41.1
Size-weighted Interconnectedness:						
Based on 2-digit Borrower SIC	270	42.4	5.8	32.2	44.1	48.7
Based on Borrower State	270	43.0	6.4	32.8	43.9	50.8
Relationship-weighted Interconnectedness:						
Based on 2-digit Borrower SIC	270	46.2	5.4	36.9	47.7	52.2
Based on Borrower State	270	45.9	5.6	36.8	46.4	52.8
Systemic Risk Measures:						
Systemic Capital Shortfall (SRISK) ($bn)	5,733	25.33	47.08	−7.29	5.96	88.72
DIP (€bn)	1343	14.70	18.51	0.51	6.54	42.15
1% CoVaR ($bn)	1,716	−15.79	31.70	−50.69	−2.56	−0.25
Lead Arranger Characteristics:						
Total Assets ($bn)	17,341	310.48	475.56	10.02	118.02	844.97
Market Share as Lead Arranger (%)	27,117	1.00	3.24	0.00	0.08	1.80
# of Loans Arranged during 12 Months	27,117	47	129	1	8	107
$ of Loans Arranged during 12 Months ($bn)	27,117	9.08	35.90	0.04	0.55	15.72
Market Characteristics:						
Market Size ($bn)	270	912.8	504.6	234.9	956.1	1,657.6

This table reports summary statistics of various distance, interconnectedness, and systemic risk measures as well as lead arranger (bank) and market characteristics. Distance between two lead arrangers is measured by their Euclidean distance as they are positioned in the Euclidean space based on their specializations in the U.S. syndicated loan market. Interconnectedness of a lead arranger is computed based on its distance from all the other lead arrangers in specializations with regard to 2-digit borrower SIC industry and borrower U.S. state and can be equal-, size-, or relationship-weighted. Market-aggregate interconnectedness is the equally weighted average of all U.S. bank's bank-level interconnectedness for each month. Systemic risk of a lead arranger is measured by SRISK, DIP, and CoVaR. We show below summary statistics of the distance measures of 2,690,674 to 2,692,389 lead arranger pair-months, the bank-level interconnectedness measures of 26,277 to 26,741 lead arranger-months, the market-aggregate interconnectedness measure of 270 month, the SRISK measures of 5733 lead arranger-months, the DIP measure of 1343 lead arranger-months, and the CoVaR measures of 1716 lead arranger-quarters. Lead arranger (bank) characteristics are reported of 27,117 lead arranger-months, and market characteristics are reported of 27 months.

two variables indicate the level of diversification and breadth of the bank's syndicated loan portfolio. While the variables diversification and number of specializations are related, there is a subtle difference: diversification considers the individual syndicated loan amounts to determine each specialization's portfolio weight, whereas the number of specializations only counts specializations independent from the loan amount.[27]

We first examine correlation between interconnectedness and each of the three variables. Then, we estimate the following multiple regression model:

$$\text{Interconnectedness}_{i,t} = \alpha + \beta_1 \cdot \text{Total Assets}_{i,t} + \beta_2 \cdot \text{Market Size}_t + \beta_3 \cdot \text{Diversification}_{i,t} + \beta_4 \cdot \text{Number of Specializations}_{i,t} + \text{Lead Arranger}_i + e_{i,t}, \quad (8)$$

where the dependent variable $\text{Interconnectedness}_{,t}$ is the level of interconnectedness of bank i in month t. $\text{Total Assets}_{i,t}$ is bank i's lagged total assets at the beginning of month t; Market Size_t is the total issuance volume in the 12 months prior to loan origination capturing a possible effect of syndicated loan market size on bank interconnectedness. For example, more lending associated with a larger syndicated loan market might mechanically increase interconnectedness

because banks are no longer are available to avoid each other in their lending specialization. $\text{Diversification}_{i,t}$ is the diversification measure computed as in Eq. (9) below:

$$\text{Diversification}_{i,t} = \left[1 - \sum_{j=1}^{J} \left(w_{i,j,t}\right)^2\right] \times 100, \quad (9)$$

where $\text{Diversification}_{i,t}$ measures the diversification level of bank i in month t and, as in (1), $w_{i,j,t}$ is the weight bank i invests in specialization j (e.g. industry) within 12 months prior to month t.[28] $\text{Number of Specializations}_{i,t}$ is the number of specializations the bank is engaged in as a lead arranger.[29] Lead Arranger_i is a vector of lead arranger (bank) fixed effects. Fahlenbrach et al. (2012) report that bank business models and risk culture are persistent and they might affect a bank's decision to become more interconnected. Standard errors are heteroscedasticity robust and clustered at the lead arranger level.

Table 3 reports the results for both specializations by borrower industry and borrower geographic location using our three types of weightings. First, we show in Panel A significantly positive Pearson correlation coefficients between interconnectedness and total assets, market size in the syndicated loan market, diversification, and number of specializations, all at the 1% level, indicating a positive association of these variables with interconnectedness. The square of the Pearson correlation coefficient helps to assess the explanatory power of these four variables for interconnectedness. We find that the size variables, total assets and market size, have Pearson correlations ranging from 0.30 to 0.40 and 0.21–0.30, respectively. This corresponds to an explanatory power of these variables for interconnectedness in an univariate setting of 9% to 16% and 4% to 10%, respectively. In contrast, diversification and number of specializations have a Pearson correlation in the range of 0.93–0.98 and 0.73–0.76, respectively. This corresponds to an explanatory power of these variables for interconnectedness in an univariate setting of 86% to 96% and 53% to 58%, respectively. In other words, banks with concentrated loan portfolios are less interconnected relative to those with diversified portfolios. Overall, diversification and number of specialization are relatively more important determinants of loan market interconnectedness than bank size and market size.

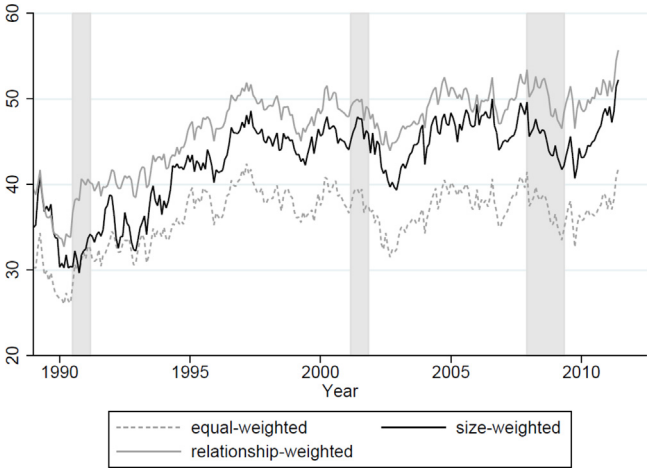


Fig. 1. Time Series of Market-Aggregate Interconnectedness, 1989–2011.
This figure shows the time series of the monthly market-aggregate Interconnectedness from January 1989 to June 2011. Interconnectedness of a lead arranger is computed based on its distance from all the other lead arrangers in specializations in the U.S. syndicated loan market. Lender specialization in this figure is based on 2-digit borrower SIC industry. The market-aggregate Interconnectedness is an equal-weighted average of all U.S. bank's bank-level interconnectedness of all the lead arrangers. Three series of market-aggregate interconnectedness are shown below, and they employ equal-, size-, and relationship-weights at the lead arranger level, respectively. Grey shaded areas highlight NBER recession periods.

We include next all variables in multivariate regressions and report the results in Table 3 Panel B. We continue to find positive effects of total assets, market size, diversification, and number of specializations on interconnectedness.[30] While the coefficients on diversification, number of specializations and market size are all significant at the 1% level, the coefficients on total assets are sometimes less or not significant in the aggregations by industry specialization. Consistent with the correlation results, diversification is a main driver of interconnectedness in all regressions.

4.3. Time trends in interconnectedness

We next investigate time-series properties of our interconnectedness measures across our three weighting schemes (i.e. equal-, size-, and relationship-weighted) over the January 1989 to June 2011 period in Figs. 1–3.[31]

At the market-aggregate level, size-, and relationship-weighted interconnectedness are consistently greater than the baseline equal-weighted counterpart throughout our sample, whereas relationship-weighted interconnectedness is consistently the largest (see Fig. 1). Higher relationship-weighted interconnectedness indicates that banks tend to establish lending relationships with those banks that have similar asset allocations in their syndicated loan portfolios. Higher size-weighted interconnectedness

[27] The Pearson correlation between both measures is about 0.7.

[28] The notion behind the measure is that as a bank becomes more diversified, $\sum_{j=1}^{J} \left(w_{i,j,t}\right)^2$ becomes smaller, so that the measure for diversification grows larger.

[29] Number of $\text{Specialization}_{i,t}$ varies by the type of specializations. For the specialization by borrower industry, it is the number of 2-digit borrower SIC industries. For the specialization by borrower geographic location, it is the number of U.S. states.

[30] The results are robust to regressions specifications without the Lead Fixed Effects, in which we include dummy variables for commercial bank, and dummies for headquarter locations for Europe, as well as outside U.S. and Europe to control for heterogeneity across lead arrangers.

[31] All figures are based on industry aggregation. Figures for regional aggregation are very similar but remain unreported for brevity.

Table 2
Effect of Distance on Likelihood of Being Chosen As A Syndicate Member.

Syndicate Member Indicator	Industry Aggregation	Regional Aggregation
Distance from Lead Arranger	−0.059*** (0.0049)	−0.051*** (0.0040)
Previous Relationship with Lead	0.020*** (0.0020)	0.022*** (0.0018)
Previous Relationship with Borrower	0.533*** (0.0105)	0.534*** (0.0105)
Market Share as a Lead	0.004*** (0.0006)	0.004*** (0.0006)
Loan Facility Fixed Effects	Yes	Yes
N=	10,887,311	10,887,313
Adjusted R^2	0.3231	0.3227

This table reports coefficient estimates from regressions relating the likelihood of a potential lender (that was among the top 100 lead arrangers in the previous twelve months) being chosen as a syndicate member by the lead arranger to the distance between the potential lender and the lead arranger. The dependent variable is an indicator variable for whether the potential lender is indeed a syndicate member. The independent variable of interest is the distance between the potential lender and the lead arranger based on their portfolios of syndicated loans originated during the previous twelve months. We use distance as an independent variable based on lender specializations in borrower industry (2-digit borrower SIC industry) and borrower region (U.S. state), respectively. Control variables include an indicator variable for whether the potential lender has previous relationship with the lead arranger, an indicator variable for whether the potential lender has previous relationship with the borrower, and the market share of the potential lender as a lead arranger in the U.S. syndicated loan market during the previous twelve months. All regressions include loan facility fixed effects. Robust standard errors allowing for clustering by lead arranger are in parentheses. * indicates that the estimated coefficient is significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

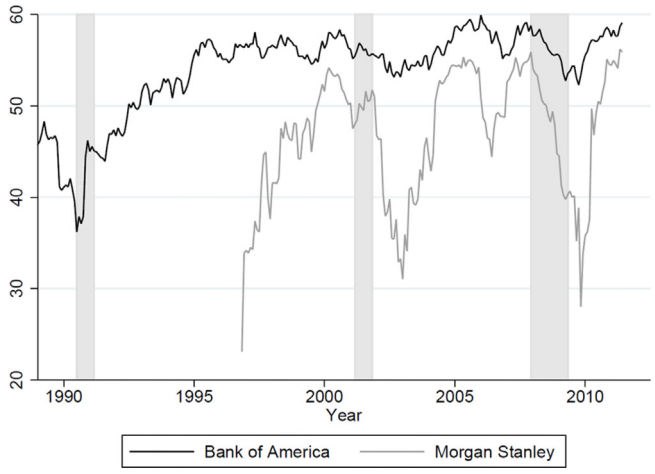


Fig. 2. Time Series of Bank-Level Interconnectedness (I), 1989–2011.
This figure shows the time series of the monthly bank-level Interconnectedness from January 1989 to June 2011. Interconnectedness of a lead arranger is computed based on its distance from all the other lead arrangers in specializations in the U.S. syndicated loan market. Lender specialization in this figure is based on 2-digit borrower SIC industry. Interconnectedness of a lead arranger is based on equal-weighting of distances to other lead arrangers. We select the time series for Bank of America and Morgan Stanley. The time series for Morgan Stanley starts in end-1996, when Morgan Stanley became active in the syndicated loan market. Grey shaded areas highlight NBER recession periods.

compared to equal-weighted interconnectedness indicates that bank's asset allocations is on average more similar to larger banks compared to smaller banks. In addition, we observe a sharp increase in the difference between the size-weighted interconnectedness and the equal-weighted interconnectedness from the early 1990s until 2007 and again after the financial crisis of 2008 and 2009 consistent with the interpretation that banks become too large and too interconnected to fail during the financial crisis (Acharya and Yorulmazer, 2008).[32] During the last two recession periods, in 2001 and 2008–2009, interconnectedness dropped significantly, but rose again following both periods.[33]

At the bank-level, we observe that banks who are frequently lead arrangers in the loan syndicated market (e.g. Bank of America) have consistently higher, and tend to have less volatile, interconnectedness measures than banks who play a smaller role in this market (e.g. Morgan Stanley), see Fig. 2. Given their importance as lead arranger, large lenders in the syndicated loan market, such as Bank of America, may not be able to specialize in certain industries or regions but originate loans across all regions and industries. In contrast, Morgan Stanley, as a medium-size lead arranger in the syndicated loan market, has more flexibility to structure its portfolio. This is demonstrated for example by the sharp drops in Morgan Stanley's interconnectedness following the last two recession periods comparted to Bank of America.[34]

Importantly, the plots generally show a fair amount of stability from year to year with our interconnectedness measure, particularly in periods of expansion.[35] This is important as the interconnectedness measure should be used by policy makers to assess the build-up of systemic risk during normal times. Wide time-series variation in interconnectedness during expansions would make the optimal policy more difficult to implement.

Fig. 3 plots the equal-weighted and size-weighted interconnectedness measure for Bank of America (Panel A) and Morgan Stanley (Panel B) as well the difference between both time-series. Consistent with Fig. 1, the plots show that size-weighted interconnectedness measure is larger compared with the equal-weighted measure and this difference is even increasing, particularly for important originators such as Bank of America. I.e. banks that play a larger or smaller role in the market tend to increase their syndicate portfolio overlap particularly with large banks consistent with the moral hazard theories mentioned above.

5. Interconnectedness and systemic risk

In this section, we investigate whether our bank-level interconnectedness measures increase bank's systemic risk (SRISK, DIP, and CoVaR) during recessions.

[32] Consistent with this interpretation, cross-sectional analyses show that the fraction of banks with the highest interconnectedness also increased prior to the financial crisis of 2008–2009.

[33] Minoiu and Reyes (2013) find the same cyclicality when studying the interconnectedness of entire banking systems across countries.

[34] For example, Morgan Stanley reduced the lending in the syndicated loan market during the recent crisis and adjusted the asset allocation. It focused lending almost entirely to two industries, namely chemical and allied products as well as electronic and other electronic equipment and components – out of mining, construction, electric, gas and sanitary services, and engineering, accounting, research and management services. Its geographical specializations primarily focused on Michigan, and also to California and New Jersey – and out of New York and Texas.

[35] We also analyze the time-series correlation of the bank-level interconnectedness measures which supports the stability of our measure. For example, the time-series correlation among the top three U.S. lead arrangers (JPMorgan Chase, Bank of America, and Citigroup) averages 0.84, and the time-series correlation among Credit Suisse and UBS is 0.82. In addition, the time-series correlations among U.S. banks and among European banks tend to be higher than between U.S. banks and European banks.

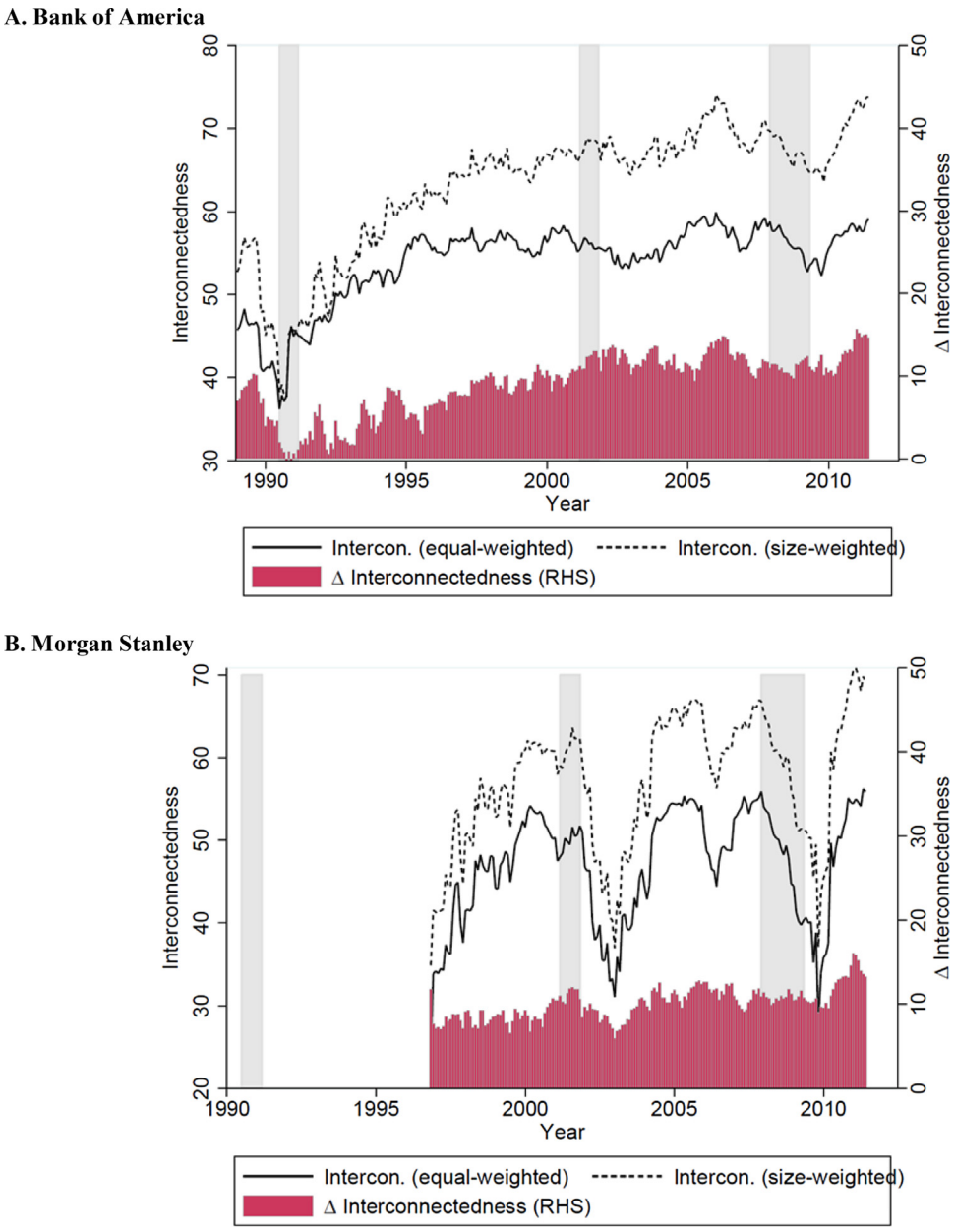


Fig. 3. Time Series of Bank-Level Interconnectedness (II), 1989–2011.
This figure shows the time series of the monthly bank-level Interconnectedness from January 1989 to June 2011. Interconnectedness of a lead arranger is computed based on its distance from all the other lead arrangers in specializations in the U.S. syndicated loan market. Lender specialization in this figure is based on 2-digit borrower SIC industry. Interconnectedness of a lead arranger is based on equal- and size-weighting of distances to other lead arrangers. We select the time series for Bank of America and Morgan Stanley. The time series for Morgan Stanley starts in end-1996, when Morgan Stanley became active in the syndicated loan market. The bars on the right hand side highlight the difference between the size-weighted and the equal-weighted bank-level interconnectedness over time. Grey shaded areas highlight NBER recession periods.

5.1. *Bank-level tests – methodology*

Interconnectedness in the syndicated loan market creates the potential for enhanced systemic risk. Banks become more vulnerable to common shocks as exposure to similar assets and banks can spread throughout the syndicate network to other banks.

To examine this, we match the bank-level systemic risk measures SRISK, CoVaR, and DIP with the time-series of our bank-level interconnectedness measure. To test their relationship, we first univariately examine the correlation between systemic risk and interconnectedness. Table 4 shows that Pearson correlation coefficients between interconnectedness and all three bank-level systemic risk measures (SRISK, DIP, and CoVaR) are significantly positive at the 1% level.

In a second step, we add control variables in a multiple regression setting. The general form of the regression we estimate is as follows:

$$\begin{aligned}\text{SystemicRisk}_{i,t} &= \alpha + \beta_1 \cdot (\text{Interconnectedness}_{i,t} \times \text{Expansion}_t) \\ &+ \beta_2 \cdot (\text{Interconnectedness}_{i,t} \times \text{Recession}_t) + \beta_3 \cdot \text{Recession}_t \\ &+ \beta_4 \cdot \text{TotalAssets}_{i,t} + \beta_5 \cdot \text{MarketShare}_{i,t} + \beta_6 \cdot \text{MarketSize}_t \\ &+ \beta_7 \cdot \text{SystemicRisk}_{i,t-1} + \text{LeadArranger}'_i + e_{i,t}. \end{aligned} \quad (10)$$

The dependent variable Systemic $\text{Risk}_{i,t}$ is the systemic risk measure of bank i in month t, which can be either SRISK, DIP, or CoVaR. The key independent variable $\text{Interconnectedness}_{i,t}$ is the

Table 3
Determinants of Interconnectedness.

A. Pearson Correlation

Pearson Correlation	N=	Industry Aggregation			Regional Aggregation		
		equal-weighted	size-weighted	relationship-weighted	equal-weighted	size-weighted	relationship-weighted
Total Assets	17,341	0.3014***	0.3402***	0.3263***	0.3782***	0.4001***	0.3871***
Market Size	26,532	0.2148***	0.2889***	0.2488***	0.2614***	0.3083***	0.2592***
Diversification	26,532	0.9772***	0.9547***	0.9450***	0.9642***	0.9415***	0.932***
# of Specializations	26,532	0.7290***	0.7345***	0.7361***	0.7599***	0.7591***	0.7602***

B. Multivariate Regressions

Bank-level Interconnectedness	Industry Aggregation			Regional Aggregation		
	equal-weighted	size-weighted	relationship-weighted	equal-weighted	size-weighted	relationship-weighted
Total Assets	−0.000 (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)	0.002* (0.000)	0.002*** (0.000)
Market Size	0.002*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.004*** (0.000)	0.002*** (0.000)
Diversification	0.362*** (0.003)	0.396*** (0.004)	0.412*** (0.005)	0.346*** (0.004)	0.382*** (0.007)	0.392*** (0.006)
# of Specializations	0.141*** (0.011)	0.236*** (0.013)	0.288*** (0.017)	0.215*** (0.020)	0.346*** (0.028)	0.376*** (0.029)
Lead Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
N=	17,341	17,341	17,248	17,341	17,341	17,248
Adjusted R^2	0.9723	0.9508	0.9524	0.9557	0.9293	0.9352

This table examines a number of bank characteristics as potential determinants of interconnectedness. Interconnectedness of a lead arranger is computed based on its distance from all the other lead arrangers in specializations with regard to 2-digit borrower SIC industry and borrower U.S. state and can be equal-, size-, or relationship-weighted. Bank characteristics include total assets (in billions of U.S. dollars), market size measured as syndicated loan originated during the previous twelve months in the U.S. syndicated loan market (in billion U.S. dollar), diversification, and the number of specializations the bank is engaged in. Panel A shows Pearson correlation coefficients between interconnectedness and bank characteristics, and Panel B reports results from multivariate regressions with lead arranger fixed effects. Robust standard errors allowing for clustering by lead arranger are in parentheses. * indicates that the estimated coefficient is significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

level of interconnectedness of bank i in month t. To distinguish between the effect of interconnectedness on systemic risk during periods of expansions and recessions, we include two interaction terms in the regression: ($Interconnectedness_{i,t} \times Expansion_t$) and ($Interconnectedness_{i,t} \times Recession_t$). $Recession_t$ ($Expansion_t$) is an indicator variable equal to one if month t falls into recessions (expansion) as measured by NBER recession dates.[36] We control for bank size ($Total\ Assets_{i,t}$), market power in loan syndication ($MarketShare_{i,t}$) and market size of the U.S. syndicated loan market ($MarketSize_t$). A one-period lagged systemic risk measure ($Systemic\ Risk_{i,t-1}$) is included on the right hand side of the regression to account for serial correlation. We further include lead arranger (bank) fixed effects. Standard errors are heteroscedasticity robust and clustered at the lead arranger level. To analyze whether interconnectedness has a stronger effect on systemic risk during recessions compared to expansions, we formally test the following hypothesis,

$$H0 : \beta_2 - \beta_1 = 0 \tag{11}$$

5.2. Interconnectedness and SRISK

Table 5 reports the regression results for SRISK based on 57 unique lead arrangers[37] with monthly data ranging from January 2000 to June 2011. In all specifications, we find statistically and economically insignificant coefficients on the interaction term interconnectedness and expansion. That is, during periods of economic expansions, interconnectedness neither elevates nor reduces SRISK. However, our empirical findings show that all coefficients on the interaction term between interconnectedness and NBER recessions are consistently positive and statistically significant at the 1% level. The effect is also economically significant: i.e. an increase of one standard deviation in interconnectedness during recessions leads to a total SRISK increase of $4.4 to $8.9 billion when the past six to twelve month were in recession, which is approximately a 17% to 35% increase from the mean SRISK. Formally testing the difference in the interaction terms between interconnectedness and recession with interconnectedness and recession consistently shows statistical significance at the 1% level.[38] These results show that interconnectedness contributes more positively to SRISK during recessions consistent with an amplifying effect of interconnectedness on systemic risk during recessions as suggested by Bernanke (2010). Our finding is also consistent with our prior as to the effect of interconnectedness on SRISK when banks are vulnerable to recessions (see Section 2.2.4).

The coefficients on a bank's total assets are significantly positive indicating that larger banks are more sensitive to systemic risk.[39] The effect of market share as a lead arranger in the syndicated loan market on SRISK is insignificant. The coefficient on market size is marginally positive and significant. The one-month lagged SRISK has significantly positive coefficients consistently around 0.9 showing high persistence of SRISK over time.[40]

[36] The NBER identifies three recession periods during our sample period: July 1990–March 1991, March 2001–November 2001, and December 2007–June 2009.

[37] Note that the number of lead arrangers slightly reduces from our matched SRISK and interconnectedness dataset described above, as the regression specification requires the total assets variable and a lagged SRISK measure. The same applies to the CoVaR and DIP regressions.

[38] The results are robust to restricting the sample to positive SRISK, that is, when financial institutions have a capital shortfall.

[39] These results are consistent with our earlier results describing the drivers of interconnectedness in corporate loan markets. While bank size is an important factor, it is not a sufficient condition that eventually explains cross-sectional variation in interconnectedness and eventually systemic risk. Recent events provide a supporting narrative. For example, the default of the Portuguese lender Banco Espirito Santo (a relatively small bank with assets worth EUR 81 billion) caused a global stock market decline in July 2014. Similarly, the Swiss regulator declared the Raiffeisenbank Schweiz Genossenschaft, a bank with assets of EUR 28 billion, "systemically important" in August 2014 because its products cannot be easily replaced and are important for the Swiss economy. In other words, systemic importance of banks extends beyond size, and it is crucial to monitor other factors such as interconnectedness of banks.

[40] We also run tests using LRMES, which is a main component of SRISK and more of a measure of tail risk, as the dependent variable. Our main results and conclusions do not change.

Table 4
Correlation between Systemic Risk and Interconnectedness.

Pearson Correlation	N=	Industry Aggregation			Regional Aggregation		
		equal-weighted	size-weighted	relationship-weighted	equal-weighted	size-weighted	relationship-weighted
SRISK	5733	0.1391***	0.1427***	0.1396***	0.2177***	0.2109***	0.2190***
DIP	1343	0.2359***	0.2470***	0.2502***	0.2406***	0.2245***	0.2496***
-1% CoVaR	1073	0.4178***	0.4246***	0.4252***	0.4502***	0.4607***	0.4594***

This table reports Pearson correlation coefficient estimates between a financial institution's systemic risk and its interconnectedness in the U.S. syndicated loan market as well as the U.S. financial sector's systemic risk and U.S. bank's market-aggregate interconnectedness in the U.S. syndicated loan market. Bank-level systemic risk is measured by systemic capital shortfall (SRISK) in billions of U.S. dollars, the monthly distress insurance premium (DIP) in billions of euros, and the opposite of 1% CoVaR in billions of U.S. dollars. Interconnectedness of a lead arranger is computed based on its distance from all the other lead arrangers in specializations with regard to 2-digit borrower SIC industry and borrower U.S. state and can be equal-, size-, or relationship-weighted. * indicates that the estimated coefficient is significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 5
Interconnectedness and SRISK.

SRISK	Industry Aggregation			Regional Aggregation		
	equal-weighted	size-weighted	relationship-weighted	equal-weighted	size-weighted	relationship-weighted
Interconnectedness× Expansion	−0.006 (0.013)	−0.006 (0.011)	−0.001 (0.011)	0.001 (0.014)	−0.001 (0.011)	0.004 (0.012)
Interconnectedness × Recession	0.061*** (0.022)	0.052*** (0.018)	0.050*** (0.018)	0.085*** (0.021)	0.072*** (0.018)	0.070*** (0.017)
Recession	−1.287 (0.784)	−1.455* (0.822)	−1.260 (0.840)	−2.146*** (0.752)	−2.343*** (0.812)	−2.149*** (0.769)
Total Assets	0.007*** (0.001)	0.007*** (0.001)	0.007*** (0.001)	0.007*** (0.001)	0.007*** (0.001)	0.007*** (0.001)
Market Share	−0.022 (0.113)	−0.022 (0.113)	-0.026n (0.113)	−0.026 (0.115)	−0.024 (0.115)	−0.030 (0.117)
Market Size	0.001* (0.000)	0.001* (0.000)	0.001* (0.000)	0.001* (0.000)	0.001* (0.000)	0.001* (0.000)
Lagged SRISK	0.901*** (0.012)	0.901*** (0.011)	0.901*** (0.011)	0.898*** (0.012)	0.899*** (0.012)	0.898*** (0.012)
Lead Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
N=	5,390	5,390	5,390	5,381	5,381	5,381
Adjusted R^2	0.9581	0.9581	0.9581	0.9591	0.9591	0.9591
H_0	0.067***	0.058***	0.051***	0.085***	0.073***	0.067***
p-value	0.003	0.002	0.005	0.000	0.000	0.000

This table reports coefficient estimates from regressions relating a financial institution's SRISK to its interconnectedness in the U.S. syndicated loan market. The dependent variable is systemic capital shortfall (SRISK) in billions of U.S. dollars. The independent variable of interest is the interconnectedness of a lead arranger, which is computed based on its distance from all the other lead arrangers in specializations with regard to 2-digit borrower SIC industry and borrower U.S. state and can be equal-, size-, or relationship-weighted. Recession is an indicator variable equal to 1 if a month falls into the recession periods identified by NBER. Expansion is an indicator variable equal to 1 if a month falls not into the recession periods identified by NBER. Interconnectedness × Expansion is the interaction term of Interconnectedness and Expansion. Interconnectedness × Recession is the interaction term of Interconnectedness and Recession. Control variables include the financial institution's total assets, market share as a lead arranger in the U.S. syndicated loan market during the previous twelve months, market size measured as syndicated loan originated during the previous twelve months in the U.S. syndicated loan market (in billion U.S. dollar), and one-month lagged SRISK. All regressions include lead arranger fixed effects. Robust standard errors allowing for clustering by lead arranger are in parentheses. The bottom part of the table shows the hypothesis test (H_0: *Interconnectedness x Recession − Interconnectedness x Expansion* = 0) and the hypothesis test's *p*-value. * indicates that the estimated coefficient is significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

5.3. Interconnectedness and DIP

Table 6 reports OLS regression results for our systemic risk measure DIP using 19 unique lead arrangers (European banks) with monthly data ranging from February 2002 to June 2011 using the same regressions specification (10) as above.

Similar to results for SRISK, we find that the coefficient of the interaction term of interconnectedness and expansion is not statistically and economically significant. Further, all coefficients of the interaction term of interconnectedness with NBER recessions are significantly positive at the 1% level. Testing our H0-hypothesis shows that these two interaction coefficients are significantly different at the 1% level, that is higher interconnectedness increases DIP during recessions, but not during normal times. This effect is also an economically significant as an increase of one standard deviation in interconnectedness during recessions is related to an increase of $7.3 to $11.1 billion in DIP when the last six to twelve month were in recession, which represents a 36% to 54% increase from the average DIP.

Table 6 also shows that a great amount of variation in DIP is explained by a bank size. The effect of market size is statistically significant, but economically small. Consistent with our SRISK results, market share is usually not statistically significant and DIP displays high persistence (around 0.75) over time.

5.4. Interconnectedness and CoVaR

Table 7 reports the multiple regression results for 1% CoVaR based on 36 unique U.S. lead arrangers with quarterly data ranging from the first quarter of 2000 until the fourth quarter of 2010. The regressions have the same specifications as above.

Results consistently show small and insignificant coefficients on the interaction term of interconnectedness and expansion for all specifications. The coefficients on the interaction term of interconnectedness and recession are consistently positive and statistically significant at the 5% or 10% level.[41] Overall, these findings show a magnifying effect of interconnectedness on CoVaR during recessions.[42] The results are also highly significant in economic terms: an increase of interconnectedness by one standard deviation during recessions results in a total increase of $3.3-4.8 billion in 1% CoVaR when the last six month were recession periods. Such increases are elevations of about 21%-30% from the average 1% CoVaR measures. The effect of interconnected-

[41] While the difference between both interaction coefficients are large in magnitude, they are not statistically significant (but almost at the 10% level) as our tests of the H0-hypothesis show.

[42] The results are robust to analyzing the 5% CoVaR instead of the 1% CoVaR.

Table 6
Interconnectedness and DIP.

DIP	Industry Aggregation			Regional Aggregation		
	equal-weighted	size-weighted	relationship-weighted	equal-weighted	size-weighted	relationship-weighted
Interconnectedness× Expansion	0.002 (0.013)	0.023 (0.015)	0.012 (0.013)	−0.001 (0.018)	0.014 (0.018)	0.005 (0.017)
Interconnectedness × Recession	0.130*** (0.038)	0.129*** (0.036)	0.105*** (0.030)	0.116*** (0.033)	0.110*** (0.031)	0.089*** (0.026)
Recession	−2.725** (1.083)	−2.629** (1.157)	−2.378** (1.047)	−2.778** (1.069)	−2.724** (1.159)	−2.303** (1.012)
Total Assets	0.004*** (0.000)	0.004*** (0.001)	0.004*** (0.000)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
Market Share	1.302 (0.751)	1.325* (0.755)	1.282 (0.750)	1.155 (0.732)	1.125 (0.738)	1.139 (0.728)
Market Size	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.001** (0.000)
Lagged DIP	0.744*** (0.028)	0.743*** (0.028)	0.745*** (0.028)	0.747*** (0.027)	0.748*** (0.027)	0.748*** (0.027)
Lead Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
N=	1,325	1,325	1,325	1,316	1,316	1,316
Adjusted R^2	0.8291	0.8292	0.8289	0.8288	0.8288	0.8286
H_0	0.128***	0.107***	0.093***	0.117***	0.096***	0.084***
p-value	0.004	0.006	0.005	0.004	0.006	0.005

This table reports coefficient estimates from regressions relating a European financial institution's DIP to its interconnectedness in the U.S. syndicated loan market. The dependent variable is the monthly distress insurance premium (DIP) in billions of euros. The independent variable of interest is the interconnectedness of a lead arranger, which is computed based on its distance from all the other lead arrangers in specializations with regard to 2-digit borrower SIC industry and borrower U.S. state and can be equal-, size-, or relationship-weighted. Recession is an indicator variable equal to 1 if a month falls into the recession periods identified by NBER. Expansion is an indicator variable equal to 1 if a month falls not into the recession periods identified by NBER. Interconnectedness × Expansion is the interaction term of Interconnectedness and Expansion. Interconnectedness × Recession is the interaction term of Interconnectedness and Recession. Control variables include the financial institution's total assets, market share as a lead arranger in the U.S. syndicated loan market during the previous twelve months, market size measured as syndicated loan originated during the previous twelve months in the U.S. syndicated loan market (in billion U.S. dollar), and one-month lagged DIP. All regressions include lead arranger fixed effects. Robust standard errors allowing for clustering by lead arranger are in parentheses. The bottom part of the table shows the hypothesis test (H_0: Interconnectedness x Recession − Interconnectedness x Expansion = 0) and the hypothesis test's *p*-value. * indicates that the estimated coefficient is significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 7
Interconnectedness and CoVaR.

-1% CoVaR	Industry Aggregation			Regional Aggregation		
	equal-weighted	size-weighted	relationship-weighted	equal-weighted	size-weighted	relationship-weighted
Interconnectedness × Expansion	0.008 (0.027)	0.001 (0.021)	0.012 (0.020)	0.022 (0.026)	0.020 (0.023)	0.021 (0.024)
Interconnectedness × Recession	0.123* (0.067)	0.098* (0.057)	0.099* (0.057)	0.152* (0.081)	0.141** (0.069)	0.125* (0.064)
Recession	−2.810 (2.209)	−2.923 (2.303)	−2.877 (2.289)	−3.320 (2.437)	−4.052 (2.741)	−3.556 (2.601)
Total Assets	−0.010 (0.006)	−0.010 (0.006)	−0.010* (0.006)	−0.010* (0.006)	−0.010* (0.006)	−0.010* (0.006)
Market Share	−0.496 (0.701)	−0.494 (0.701)	−0.506 (0.703)	−0.505 (0.704)	−0.498 (0.701)	−0.513 (0.703)
Market Size	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)
Lagged CoVaR	0.519*** (0.081)	0.519*** (0.081)	0.519*** (0.081)	0.516*** (0.082)	0.514*** (0.082)	0.516*** (0.082)
Lead Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
N=	1035	1035	1035	1029	1029	1029
Adjusted R^2	0.3177	0.3176	0.3175	0.3191	0.3195	0.3188
H_0	0.114	0.097	0.088	0.130	0.122	0.105
p-value	0.148	0.151	0.152	0.134	0.114	0.135

This table reports coefficient estimates from regressions relating a U.S. financial institution's CoVaR to its interconnectedness in the U.S. syndicated loan market. The dependent variable is the opposite of 1% CoVaR in billions of U.S. dollars. The independent variable of interest is the interconnectedness of a lead arranger, which is computed based on its distance from all the other lead arrangers in specializations with regard to 2-digit borrower SIC industry and borrower U.S. state and can be equal-, size-, or relationship-weighted. Recession is an indicator variable equal to 1 if a month falls into the recession periods identified by NBER. Expansion is an indicator variable equal to 1 if a month falls not into the recession periods identified by NBER. Interconnectedness × Expansion is the interaction term of Interconnectedness and Expansion. Interconnectedness × Recession is the interaction term of Interconnectedness and Recession. Control variables include the financial institution's total assets, market share as a lead arranger in the U.S. syndicated loan market during the previous twelve months, market size measured as syndicated loan originated during the previous twelve months in the U.S. syndicated loan market (in billion U.S. dollar), and one-quarter lagged CoVaR. All regressions include lead arranger fixed effects. Robust standard errors allowing for clustering by lead arranger are in parentheses. The bottom part of the table shows the hypothesis test (H_0: *Interconnectedness x Recession – Interconnectedness x Expansion* = 0) and the hypothesis test's *p*-value. * indicates that the estimated coefficient is significantly different from zero at the 10% level, ** at the 5% level, and *** at the 1% level.

ness on CoVaR during recessions is smaller as compared to, e.g. SRISK or DIP. As described above, a possible interpretation is that the marginal impact of an increase in individual bank risk on the market is smaller when the economy is already in a recession.

As mentioned in Section 2.2, CoVaR is defined such that it is not explicitly sensitive to size. Consistently, we see insignificant coefficients on a bank's total assets in the regression results for CoVaR, which include bank fixed effects. A bank's market share in the syndicated loan market has no measurable effect on CoVaR, either. Also the coefficient for market size is marginal and not statistically significant. Persistence in CoVaR is indicated by the highly significant and positive coefficients (around 0.5) on the one-quarter lagged CoVaR.

5.5. Exposure versus origination

Our distance and interconnectedness measures are constructed based on loan origination during the prior 12 months. An alternative approach to calculate our interconnectedness measure is to calculate the portfolio overlap between any two banks using originations in the prior 48 months, which is the average maturity.

However, increasing the time-period over which we construct our interconnectedness measure might not increase the precision of our estimates. First, we construct the measure based on the mean/median maturity of syndicated loans but do not have information about changes in the maturity after origination (e.g. early repayment). Second, we do not have information related to changes

in exposures after origination, for example, due to loan sales or hedging of exposures that we cannot observe. Moreover, from our analysis of the organizational structure in syndicated loans we know that financial institutions participate in syndicated loans with other financial institutions that have similar syndicated loan portfolios and thus repeatedly collaborate in syndications over time. Thus, we do not expect that the approximation of the syndicated loan exposures using 48 months differ substantially from our measure of syndicated loan origination.

As a robustness test, we construct our interconnectedness measure using the prior 48 month. We find that the correlation between this and the original measure using loan originations over the prior 12 months is more than 80% using both industry and regional aggregation. We then repeat all regressions for SRISK, DIP and CoVaR using the new measure. The results are very similar. Thus, loan originations in the prior 12 months is a good approximation of a bank's portfolio decision over the prior 48 months.

6. Conclusions

Loan syndication increases the overlap of bank loan portfolios and makes them more vulnerable to contagious effects. While banks seem to diversify by syndicating loans to other banks, it increases systemic risk of the financial system, because banks become more similar to one another. To measure banks' interconnectedness in the form of common exposures, we develop a novel measure of loan market interconnectedness. Interconnectedness is driven mainly by bank diversification, less by bank size or overall loan market size. Using different market based measures of systemic risk, we find that interconnectedness is positively correlated with these systemic risk measures, and that such a positive correlation mainly arises from an elevated effect of interconnectedness on systemic risk during recessions.

Our results have several important implications for banks and regulators. First, market based measures are informative during bad times, because they pick up fundamental risks of banks precisely in a moment when banks are worried about their counterparties' exposures.

Second, we provide an important link from balance sheet risk to market-based risk measures, i.e. common exposures to large syndicated loans. This is important for regulators. Knowing that common exposures to large corporate loans are an important contributor to systemic risk helps regulators to monitor the build-up of risk in the system. We provide a first step in the quantification of these exposures on the asset side of the balance sheet. Regulators with more detailed data can extend our analyses investigating and monitoring specific industry overlap, common exposures to leveraged loans or, for example, exchange rate risks that might be hidden in these loans.[43]

Third, an institution-oriented approach to assessing and limiting systemic risk exposure is insufficient as the narrative of the recent financial crises suggests. Banks do not internalize the risks they create for the financial system as a whole. The Bank of International Settlement (BIS) published an updated methodology to identify "Global Systemically Important Financial Institutions" (G-SIFIs) in July 2013 (BIS, 2013). The indicators to identify G-SIFIs comprise five factors: (1) bank size, (2) interconnectedness, (3) substitutability of services, (4) complexity, and (5) cross-border activity, each with an equal weight. While these factors include interconnectedness, its level is determined by a fixed 20% weight shared between aggregate liability and asset interconnectedness. We propose an additional commonality measure using large corporate loans as an additional indicator helping to identify G-SIFIS.

Fourth, the Financial Stability Oversight Council (FSOC), which was created in the U.S. following the Dodd-Frank Wall Street Reform after the 2008–2009 financial crisis, has the mandate to monitor and address the overall risks to financial stability. It has the authority to make recommendations as to stricter regulatory standards for the largest and most interconnected institutions to their primary regulators. We propose using interconnectedness through large corporate loans as part of FSOC's systemic risk oversight and monitoring system.

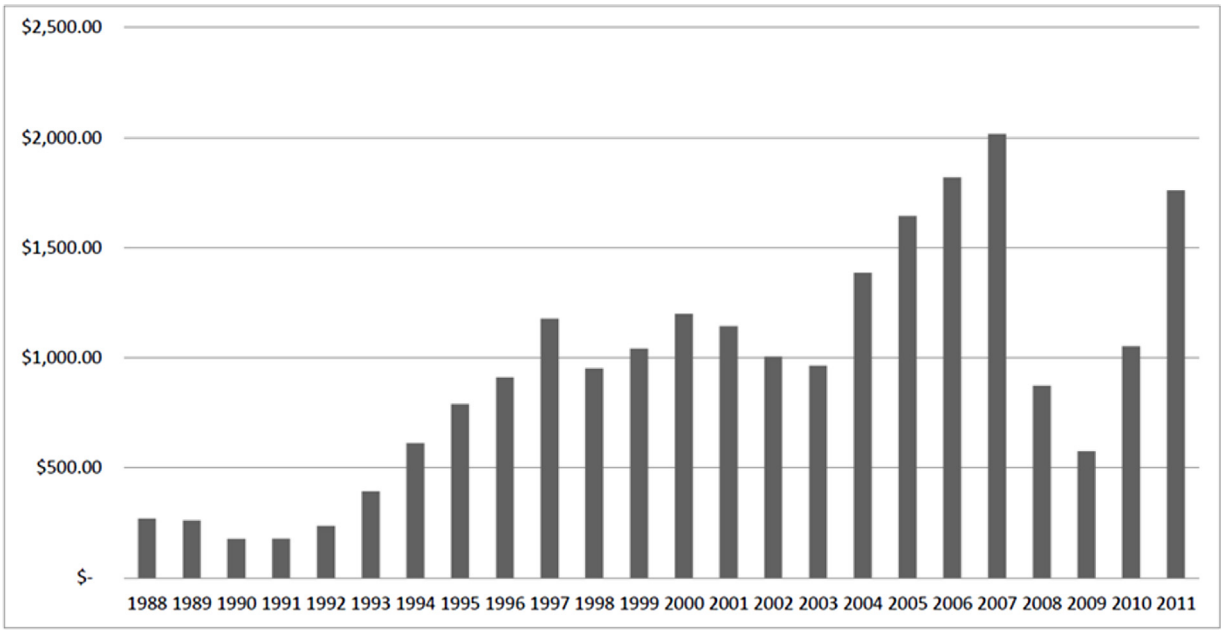
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Appendix A. The U.S. Syndicated Loan Market, 1988–2011

This Appendix shows the size of the U.S. syndicated loan market by year from 1988 to 2011. Market size is measured by the total newly originated syndicated loan amount during the year in billions of U.S. dollars. Note that data for the year of 2011 is linearly projected based on the originated amount through June of that year.

[43] The Thai financial crisis of 1997–1998 illustrates this. International banks made loans in U.S. dollar to Thai banks and these, in turn, lent to Thai firms in U.S. dollar to eliminate the exchange rate risks. After the devaluation of the Baht against the dollar, firms could not repay their U.S. dollar denominated debt and the Thai banks started to default on foreign lenders. Before the crisis, the exposure to Thai banks was identified as credit risk and the, at hindsight more important, (correlated) exposure to the Baht remained hidden.



Appendix B. Examples of Computing Distance between Lead Arrangers

This Appendix shows how distance is computed by examples. Distance between two lead arrangers is measured by their Euclidean distance as they are positioned in the Euclidean space based on their specializations in the U.S. syndicated loan market. Lender specialization in this Appendix is based on borrower SIC industry division. We show below the computation of such distance among JPMorgan Chase (JPM), Bank of America (BAC), and Citigroup (C), which were the top three lead arrangers as of January 2007 based on their portfolios of syndicated loans originated during the previous twelve months. Note that distance is the key component for computing interconnectedness – the smaller the distance between two lead arrangers, the more interconnected they are.

SIC Industry Division (2-digit SIC Industries)	JPM (1st)	BAC (2nd)	C (3rd)	(JPM-BAC)^2	(JPM-C)^2	(BAC-C)^2
Agriculture, Forestry & Fishing (01–09)	0.03%	0.17%	0.00%	0.00000198	0.00000008	0.00000287-
Mining (10–14)	5.10%	3.75%	4.77%	0.00018203	0.00001054	0.00010498
Construction (15–17)	2.34%	6.35%	0.31%	0.00160872	0.00041275	0.00365121
Manufacturing (20–39)	28.69%	23.35%	35.30%	0.00284811	0.00437535	0.01428362
Transportation, Communications, Electric, Gas & Sanitary Services (40–49)	12.30%	12.02%	20.12%	0.00000753	0.00612126	0.00655812
Wholesale Trade (50–51)	2.46%	3.82%	0.90%	0.00018570	0.00024177	0.00085125
Retail Trade (52–59)	6.81%	7.36%	2.83%	0.00003013	0.00159001	0.00205790
Finance, Insurance & Real Estate (60–67)	29.18%	30.71%	18.48%	0.00023371	0.01145803	0.01496453
Services (70–89)	13.09%	12.44%	17.18%	0.00004280	0.00166749	0.00224458
Public Administration (91–97)	0.00%	0.02%	0.11%	0.00000005	0.00000120	0.00000076
Total	100%	100%	100%	0.00514075	0.02587848	0.04471983
			Distance:	0.07169905	0.16086790	0.21147063

Note that distance is the key component for computing interconnectedness – the smaller the distance between two lead arrangers, the more interconnected they are.

	JPM	BAC	C	WB	CSGN	DB	RBS	GS	BARC	UBSN
JPM	–									
BAC	0.07	–								
C	0.16	0.21	–							
WB	0.23	0.21	0.24	–						
CSGN	0.34	0.35	0.28	0.32	–					
DB	0.17	0.19	0.14	0.17	0.28	–				
RBS	0.30	0.34	0.19	0.29	0.30	0.20	–			
GS	0.25	0.28	0.13	0.19	0.26	0.16	0.18	–		
BARC	0.44	0.45	0.35	0.28	0.43	0.36	0.38	0.24	–	
UBSN	0.41	0.42	0.39	0.41	0.17	0.41	0.43	0.37	0.52	–

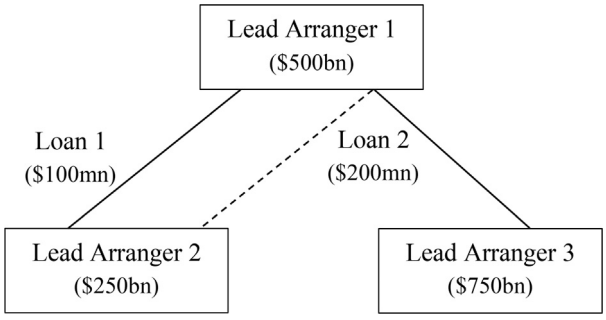
Appendix C. Distance among Top Ten Lead Arrangers

This Appendix shows distance between any two top ten lead arrangers as of January 2007 based on their portfolios of syndicated loans originated during the previous twelve months. Distance between two lead arrangers is measured by their Euclidean distance as they are positioned in the Euclidean space based on their specializations in the U.S. syndicated loan market. Lender specialization in this Appendix is based on borrower SIC industry division. The top ten lead arrangers as of January 2007 were: JPMorgan Chase (JPM), Bank of America (BAC), Citigroup (C), Wachovia Bank (WB), Credit Suisse (CSGN), Deutsche Bank (DB), Royal Bank of Scotland (RBS), Goldman Sachs (GS), Barclays (BARC), and UBS (UBSN).

Appendix D. Example of Computing Weights among Lead Arrangers

This Appendix shows how weights among lead arrangers in our bank-level interconnectedness measure are computed by an illustrative example. We use three kinds of weighting schemes: First, we assign equal weights to all other lead arrangers ("equal-weighted interconnectedness"). Second, we weight by size using a ratio of lagged total assets of lead arranger k over the sum of lagged total assets by all other lead arrangers ("size-weighted interconnectedness"). Third, we weight by contractual relationships based on the number of collaborative relationships of lead arranger i and lead arranger k relative to the total number of relationships

of lead arranger i with all other lead arrangers ("relationship-weighted interconnectedness"). We show below an illustrative example comprising of three lead arrangers and two syndicated loans. Lead arranger 1 and 2 are leads in loan 1, which has a loan amount of $100mio. Lead arranger 1 and 3 are leads in loan 2, which has a loan amount of $200mio. In addition, lead arranger 2 is participant in loan 2. Lagged total assets for each lead arranger are in parenthesis .



The computations of lead arranger 1's weights ($x_{1,2,t}$ and $x_{1,3,t}$) in the computation of bank-level interconnectedness according to formula 2 (also shown below) are as follows:

$$\text{Interconnectedness}_{i,t} = \left(1 - \sum_{k \neq i} x_{i,k,t} \cdot \text{Distance}_{i,k,t}\right) \times 100 \quad (2)$$

	$x_{1,2,t}$	$x_{1,3,t}$
Equal-weighted:	½ (=1/N, with N = 2)	½ (=1/N, with N = 2)
Size-weighted:	¼ (=$250bn/($250bn + $750bn))	¾ (=$750bn/($250bn + $750bn))
Relationship-weighted:	2/3 (=2/(2 + 1))	1/3 (=1/(2 + 1))

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